

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTEEC402 - POWER SYSTEM

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. What is the principle of operation of steam power plant?
a) Carnot cycle b) Brayton cycle c) Stirling cycle d) Rankine cycle
2. What is the effect of rise in temperature on sag when Ice and wind effect are eliminated?
a) Sag decreases b) Sag increases
c) Sag remains constant d) Sag becomes zero
3. An overhead line has three units in string of insulators. If the voltage across top unit, middle unit and bottom unit are 6 KV, 8KV and 10KV, then find the string efficiency
a) 80% b) 70% c) 60% d) 90%
4. A low load factor can lead to:
a) Lower cost per unit of energy b) Higher cost per unit of energy
c) Efficient power plant operation d) Reduced peak demand
5. If the radius of the conductor in a single-phase line is increased, what happens to the GMR?
a) GMR remains the same. b) GMR increases.
c) GMR decreases. d) GMR becomes zero
6. What is the value of the charging current flowing to earth in medium transmission line.
a) Very high b) Medium c) Negligible d) No capacitance
7. What is the main advantage of using ACSR conductors in transmission lines?
a) Lower cost b) Reduced sag
c) Increased voltage drop d) Higher current carrying capacity
8. What happens to the alternator's output voltage if the exciter field current is increased?
a) It decreases b) It increases c) It remains constant d) It fluctuates randomly
9. If the capacitance between two conductors of 3 phase line is 6 μF , the capacitance of each conductor to neutral will be
a) 3 μF b) 12 μF c) 6 μF d) 1.5 μF

- 10 For a lagging power factor load, the voltage regulation of a transmission line is typically:
a) Positive. b) Negative. c) Zero. d) Can be either positive or negative
- 11 In a bundled conductor system, what is the primary benefit of using multiple conductors per phase?
a) Increased line capacitance. b) Reduced line inductance. c) Reduced voltage drop. d) Reduced corona loss
- 12 Transposition of transmission line is done to
a) reduce line loss b) balance voltage drop
c) reduce skin effect d) reduce corona effect.
- Q2. Solve the following.
- A) With the help of block diagram explain the main components of Thermal power Plant. 6
- B) Explain the working of Hydro power plant using a neat diagram. Also state the advantages & disadvantages of it. 6
- Q3. Solve the following.
- A) State & explain the factors affecting corona loss & also methods to reduce corona loss. 6
- B) Discuss the following in detail a. Skin effect b. Ferranti effect c. Proximity effect 6
- Q4. Solve Any Two of the following.
- A) A 132 KV transmission line has following data, weight of conductor = 680kg/km; Length of span = 260m; Ultimate strength = 3100kg; Safety factor = 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 meters. 6
- B) Define String efficiency & discuss the methods to improve the string efficiency. 6
- C) State & explain the different types of insulators used in overhead transmission line 6
- Q5. Solve Any Two of the following.
- A) State the various classification of distribution system and explain any one. 6
- B) A single phase transmission line transmits 500KW, 10 KV at 0.85 lagging power factor. It has total loop resistance of $2\ \Omega$ & inductive reactance of $3\ \Omega$. Determine Voltage regulation & transmission efficiency. 6
- C) Explain the major electrical equipment's used in generating stations. 6
- Q6. Solve Any Two of the following.
- A) Write advantages & disadvantages of DC transmission system over Ac transmission system. 6
- B) What are the different types of medium transmission line? Discuss any one with the help of circuit & phasor diagram 6
- C) Explain the effect of earth on capacitance of conductor. 6

*** End ***